Ryosuke Shimada

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Education

- Doctor, Mathematical Sciences, the University of Tokyo, April 2021-March 2024.
  Thesis: Geometric Structure of Affine Deligne-Lusztig Varieties for GL_n
  Advisor: Yoichi Mieda
- Master, Mathematical Sciences, the University of Tokyo, April 2019-March 2021.
  Thesis: Geometric Structure of Affine Deligne-Lusztig Varieties for GL_3
  Advisor: Yoichi Mieda
- Bachelor, Mathematics, the University of Tokyo, April 2015-March 2019.

Academic Positions

- Program-Specific Assistant Professor, Hakubi center/Department of Mathematics, Kyoto University, April 2025-present.
- Visiting Scholar, Department of Mathematics, the University of California, Berkeley, April 2025-present (Host: Sug Woo Shin).
- Postdoctoral Fellow, Department of Mathematics, the University of Hong Kong, May 2024-March 2025 (Mentor: Xuhua He).
- Project Researcher, Graduate School of Mathematical Sciences, the University of Tokyo, April 2024-May 2024 (Mentor: Naoki Imai).

Research Interests

Affine Deligne-Lusztig Varieties, Affine Lusztig Varieties, Affine Springer Fibers, Shimura Varieties, Langlands Program.

Grants/Fellowships

- JSPS KAKENHI, Grant-in-Aid for Early-Career Scientists (Grant No. JP26K16963), April 2026-March 2031.
- JSPS KAKENHI, Grant-in-Aid for Research Activity Start-up (Grant No. JP25K23334), July 2025-March 2027.

・JSPS Overseas Research Fellowships, April 2025-March 2027.
・Basic Science Research Projects, the Sumitomo Foundation, November 2024-November 2025.
・JSPS Overseas Challenge Program for Young Researchers, August-November 2023.
・JSPS Research Fellow (DC1), Japan Society for the Promotion of Science (Grant number: JP21J22427), April 2021-March 2024.
・WINGS-FMSP Course Student, World-leading Innovative Graduate Study for Frontiers of Mathematical Sciences and Physics, October 2019-March 2024.
・Study and Visit Abroad Program (SVAP), the Faculty of Science, the University of Tokyo, January 2019-February 2019.

Awards

・Dean's Award (Doctoral program), Graduate School of Mathematical Sciences, the University of Tokyo, March 2024.
・Dean's Award (Master's program), Graduate School of Mathematical Sciences, the University of Tokyo, March 2021.

Scholarships

・The Julius Springer Scholarship, April 2021-March 2024.
・The Daiohs Foundation Scholarship, April 2021-March 2024.
・The Dentsu Scholarship, April 2015-March 2021.

Academic Visits

・The University of California, Berkeley (S. W. Shin), April 2025-present.
・The University of Duisburg-Essen (U. Görtz), August-November 2023.
・The Chinese University of Hong Kong (X. He), June 2023.
・The University of California, Berkeley (S. W. Shin), February 2023.
・The University of Maryland (T. Haines), November 2022.
・The Technical University of Darmstadt (T. Richarz), October 2022.
・The 2022 Summer School on the Langlands program, IHES, July 2022.
・The University of Chicago (K. Kato), January 2019-February 2019.

Teaching Experiences

・Instructor, *Senior mathematics seminar,* the University of Hong Kong, 2nd Semester 2024-2025.
・RA, Computer and Network Management Group, the University of Tokyo, August 2021-

March 2024.

・TA, Mathematics Help Room, the University of Tokyo, Spring 2022.

・TA, *Introduction to Mathematical Science*, the University of Tokyo, Autumn 2021.

・TA, *Algebra I/ Exercise in Algebra I*, the University of Tokyo, Spring 2021.

・TA, *Exercise in Linear Algebra,* the University of Tokyo, Autumn 2020.

・TA, *Algebra I/ Exercise in Algebra I*, the University of Tokyo, Spring 2020.

Papers

[1] R. Shimada, Regular loci of affine Springer fibers at very special parahoric level, arXiv: 2611.13601 (2026), submitted.

[2] R. Shimada, The number of Ekedahl-Oort strata intersecting the supersingular locus in the Siegel modular variety, arXiv: 2609.27198 (2026), submitted.

[3] R. Shimada and T. Takamatsu, Some generalizations of Oort's conjecture, arXiv: 2609.03188 (2026), submitted.

[4] R. Shimada, On J-strata with Parahoric Stabilizers in Affine Deligne-Lusztig Varieties, arXiv: 2606.03062 (2026), submitted.

[5] R. Shimada, On a group-theoretic approach to the supersingular locus of Shimura varieties, to appear in RIMS Kôkyûroku Bessatsu.

[6] R. Shimada, On the supersingular locus of the GU(2, n-2) Shimura variety, arXiv: 2410.05110 (2024), submitted.

[7] R. Shimada and T. Takamatsu, On the supersingular locus of the Siegel modular variety of genus 3 or 4, arXiv: 2403.19505 (2024), to appear in Ann. Inst. Fourier (Grenoble).

[8] R. Shimada, Basic loci of positive Coxeter type for GL_n, arXiv: 2402.13216 (2024), submitted.

[9] F. Schremmer, R. Shimada and Q. Yu, Affine Deligne-Lusztig Varieties of Positive Coxeter Type, arXiv: 2312.02630 (2023), submitted.

[10] R. Shimada, ピンホールモデルにおける再構成問題と箙多様体, 数理科学実践研究レター2023, LMSR 2023-13.

[11] R. Shimada, The Ekedahl-Oort stratification and the semi-module stratification, Canad. J. Math. 78 (2026), no. 3, 695–732.

[12] R. Shimada, Semi-modules and crystal bases via affine Deligne-Lusztig varieties, Adv. Math. 441(2024), Paper no. 109565.

[13] R. Shimada, On some simple geometric structure of affine Deligne-Lusztig varieties for GL_n, Manuscripta Math. 173(2024), no.3-4, 977-1001.

[14] R. Shimada, Geometric structure of affine Deligne-Lusztig varieties for GL_3, J. Algebra 623 (2023), 86-126.

Invited Talks in Seminars

[1] On J-strata in affine Deligne-Lusztig varieties, Number Theory Seminar, Kyoto University, June 12, 2026.

[2] Beyond the Chen-Zhu conjecture, NCTS Number Theory Seminar, National Center for Theoretical Sciences, January 30, 2026.

[3] Beyond the Chen-Zhu conjecture, RTG Arithmetic Geometry, Number Theory and Representation theory Seminar, University of California, Berkeley, December 1, 2025.

[4] Beyond the Chen-Zhu conjecture, The 2nd MMS Workshop for Young Researchers, Kyoto University, November 21, 2025.

[5] Shimura varieties and Kashiwara crystal, Combinatorics summer school, Kamenoi Hotel, September 30, 2025.

[6] Introduction to affine Deligne-Lusztig varieties, YNU Algebra Seminar, Yokohama National University, May 21, 2025.

[7] On the Chen-Zhu conjecture, Workshop on Affine Deligne-Lusztig varieties and Related topics, SCMS, March 6, 2025.

[8] On the Supersingular Locus of the GU(2, n-2) Shimura Variety, Algebraic Number Theory and Related Topics 2024, RIMS, January 9, 2025.

[9] On the Supersingular Locus of the GU(2, n-2) Shimura Variety, Workshop on arithmetic geometry, the University of Hong Kong, December 5, 2024.

[10] On a group-theoretic approach to the supersingular locus of Shimura varieties, Theory and Applications of Supersingular Curves and Supersingular Abelian Varieties II, RIMS, November 8, 2024.

[11] Basic loci of positive Coxeter type, the 23rd Sendai-Hiroshima Workshop on Number Theory, Tohoku University, July 9, 2024.

[12] Beyond the cases of Coxeter type, Mathematical Society of Japan Spring Meeting 2024, Osaka Metropolitan University, March 18, 2024.

[13] Beyond the cases of Coxeter type, Mittagsseminar zur Arithmetik, the University of Münster, November 7, 2023.

[14] Beyond the cases of Coxeter type, Research Training Group Seminar, the University of Duisburg-Essen, October 26, 2023.

[15] Crystal bases and affine Deligne-Lusztig varieties, the 22th Hiroshima-Sendai Workshop on Number Theory, Hiroshima University, July 14, 2023.

[16] Crystal bases and affine Deligne-Lusztig varieties, Algebraic Lie Theory and Representation Theory 2023, Tokyo Institute of Technology, May 14, 2023.

[17] Semi-modules and crystal bases via affine Deligne-Lusztig varieties, the 19th

Mathematics Conference for Young Researchers, Hokkaido University, March 7, 2023.

[18] Semi-modules and crystal bases via affine Deligne-Lusztig varieties, Berkeley Number Theory Colloquium, University of California, Berkeley, February 15, 2023.

[19] Semi-modules and crystal bases via affine Deligne-Lusztig varieties, Lie Groups and Representation Theory Seminar, the University of Maryland, November 16, 2022.

[20] Semi-modules and crystal bases via affine Deligne-Lusztig varieties, GAUS-Seminar, Technischen Universität Darmstadt, October 20, 2022.

[21] Semi-modules and crystal bases via affine Deligne-Lusztig varieties, Number Theory Seminar, Hokkaido University, September 21, 2022.

[22] On some simple geometric structure of affine Deligne-Lusztig varieties for GL_n, Mathematical Society of Japan Autumn Meeting 2022, Hokkaido University, September 15, 2022.

[23] On some simple geometric structure of affine Deligne-Lusztig varieties for GL_n, Number Theory Seminar, Kyoto University, May 27, 2022.

[24] Geometric structure of affine Deligne-Lusztig varieties for GL_3, Mathematical Society of Japan Spring Meeting 2022, Saitama University(online), March 31, 2022.

[25] Geometric structure of affine Deligne-Lusztig varieties, the 18th Mathematics Conference for Young Researchers, Hokkaido University (online), March 3, 2022.

[26] Geometric structure of affine Deligne-Lusztig varieties for GL_3, the 20th Hiroshima-Sendai Workshop on Number Theory, Hiroshima University (online), July 13, 2021.

[27] Geometric structure of affine Deligne-Lusztig varieties for GL_3, Number Theory Seminar, the University of Tokyo (online), May 26, 2021.